

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov/ Dec -2024

AI in Cyber Security [CSE-4027]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Imagine you are part of a cybersecurity team for a financial institution where recent phishing attacks have increased. Your task is to develop a knowledge-based system that can help in early detection and response to such attacks.
 - A) Question: Analyze the steps needed to implement knowledge engineering principles in developing a phishing detection system. How would you ensure this system evolves with emerging attack patterns? (5)
 - B) Scenario: A law enforcement agency is using a digital forensics ontology to assist in investigating a recent data breach incident. The ontology helps organize evidence types, investigation steps, and potential outcomes.
 - a. Question: Analyze how a digital forensics ontology enhances the investigation process for cybersecurity incidents. What challenges might arise when attempting to apply this ontology to a new type of cybercrime, and how could they be addressed? (5)
- 2) A cybersecurity firm is developing a knowledge representation model for cyber-situational awareness using ontologies to encode network elements, threats, and responses.
 - A) Question: Analyze the advantages of using ontologies to formally represent network knowledge for cyber-situational awareness. What are the potential challenges in maintaining an up-to-date ontology with rapidly changing cyber threats, and how might these be addressed? (5)
 - B) A cybersecurity monitoring system encounters vague or ambiguous data related to potential threats, such as non-specific error messages or incomplete intrusion indicators. Question: Assess how data vagueness complicates cyber-situational awareness efforts. What strategies could be used to interpret vague data for more accurate threat assessment, and what are the limitations of these approaches? (5)
- 3) A machine learning engineer is tasked with defending a recommendation system from poisoning attacks aimed at degrading its accuracy and user trust.
 - A) Evaluate the strategies available for defending against poisoning attacks in machine learning. Which methods are most effective in preserving model accuracy and integrity, and what are the limitations of these defense mechanisms? (5)
 - B) A company has implemented an exploit analysis model to assess vulnerabilities in its software products, considering factors such as the likelihood of exploitation, timing of past exploits, vendor, and programming language. (5)

Question: Assess how time-based, vendor-/platform-based, and language-based analysis contribute to understanding the likelihood of a vulnerability being exploited. How would these analyses help prioritize patching efforts, and what are the limitations of each approach?

- 4) A security team is deploying various binary classifiers, such as neural networks and support vector machines, to detect potential network attacks in real-time.
- A) Question: Analyze the effectiveness of neural networks, neuro-fuzzy networks, and support vector machines as binary classifiers in detecting network attacks. How do these classifiers differ in handling false positives and false negatives, and what implications does this have for real-world applications? (5)
- B) A cybersecurity team is evaluating different deployment methods (e.g., host-based, network-based) and detection methodologies (e.g., signature-based, anomaly-based) for a network intrusion detection system.
- Question: Analyze the pros and cons of different NIDS deployment methods and detection methodologies. How could each approach influence detection accuracy and system performance, and what factors should guide the choice between host-based and network-based intrusion detection? (5)
- 5) A cybersecurity firm is comparing the performance of two approaches, Parameterizing and Extending DroidRisk, to identify malware within Android applications.
- A) Question: Compare the effectiveness of Parameterizing and Extending DroidRisk for Android malware detection. What are the main differences in their approach to malware identification, and how do these differences impact their accuracy and generalizability in detecting new types of malware? (5)
- B) A cybersecurity researcher is preparing a novel dataset for Android malware detection, analyzing APK files and encoding features based on metadata and label assignment.
- Question: Examine the importance of application metadata, accurate label assignment, and data encoding in preparing datasets for Android malware detection. How could mislabeling or improper encoding affect the model's accuracy, and what steps can ensure dataset reliability? (5)

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